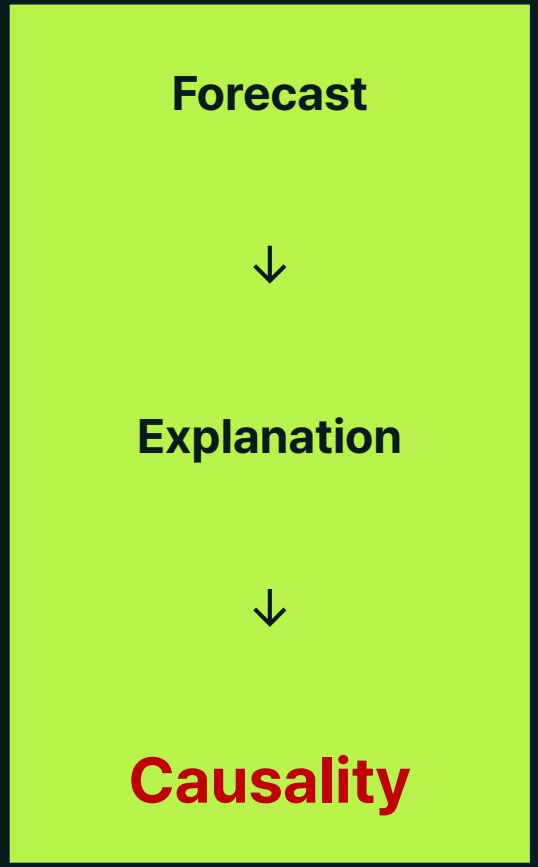
Causality · XAI · Fairness
Questions Before Trust

Prediction, explanation and cause are different



Financial ML · Week 14

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Correlation vs. causation, Shapley** Confounders, sharing credit in games
02	**Core concepts**	**Three questions** Confounding, SHAP, group metrics
03	**Cases**	**Hidden by averages** Berkeley admissions, hiring tool, card limits
04	**Summary**	**Evidence fit for the claim** Check Q&A

Prediction is not intervention effect

01

Basics

Correlation, causation, explanation

Correlation is not causation

Moving together does not show what changes what

CORRELATION

Correlation

- How much two variables move together.

CAUSATION

Causal?

- Changing one changes the other.

CONFOUNDER

Confounder

- A third variable affecting both.

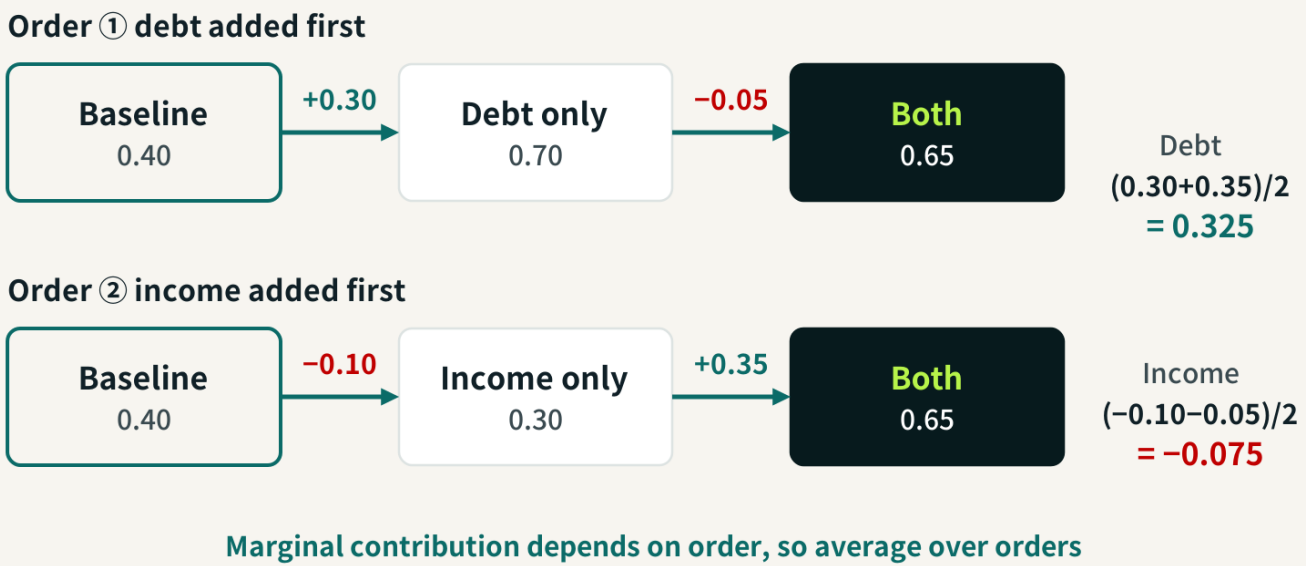
Umbrella sales predict rain, but selling umbrellas won't make it rain

Shapley values come from sharing credit in cooperative games

Lloyd Shapley (1953): calculation order via the debt and income example

Origin

- Splitting a jointly earned payoff.
- Lundberg and Lee (2017) brought it to ML as SHAP.



Average added contribution over orders

SHAP's fairness is a math property, not social fairness

02

Predict · Explain · Cause

Source concepts: confounding, experiments, SHAP, group metrics

Prediction, explanation and causality are separate questions

Umbrella sales predict rain, but selling umbrellas cannot make rain

Question	What it finds	Approach
What will happen?	**Future outcomes**	**Forecast model**
Why did the model say that?	**Feature contributions**	**SHAP and similar**
If I change it, does Y change?	**Intervention effect**	**Experiments, causal inf.**

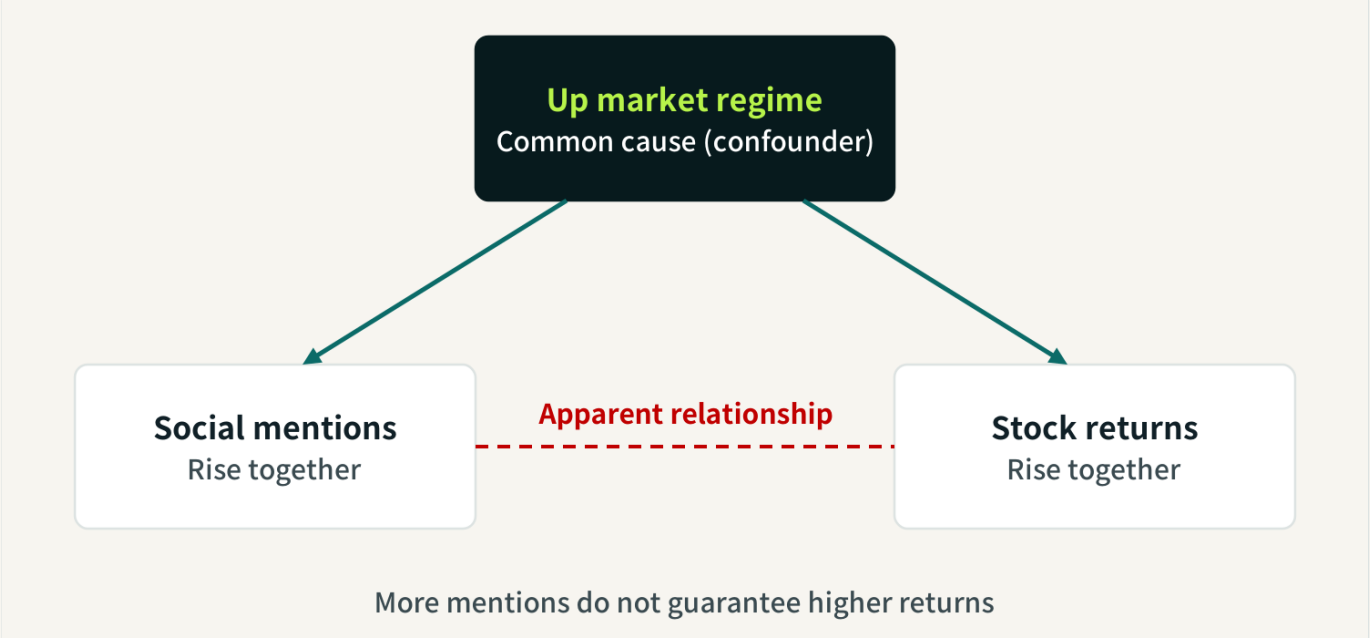
Prediction is not intervention effect

Two co-moving variables may share a common cause

A common cause is called a confounder

E.g.

- A bull market lifted both social mentions and returns.
- Mentions and returns share one cause.



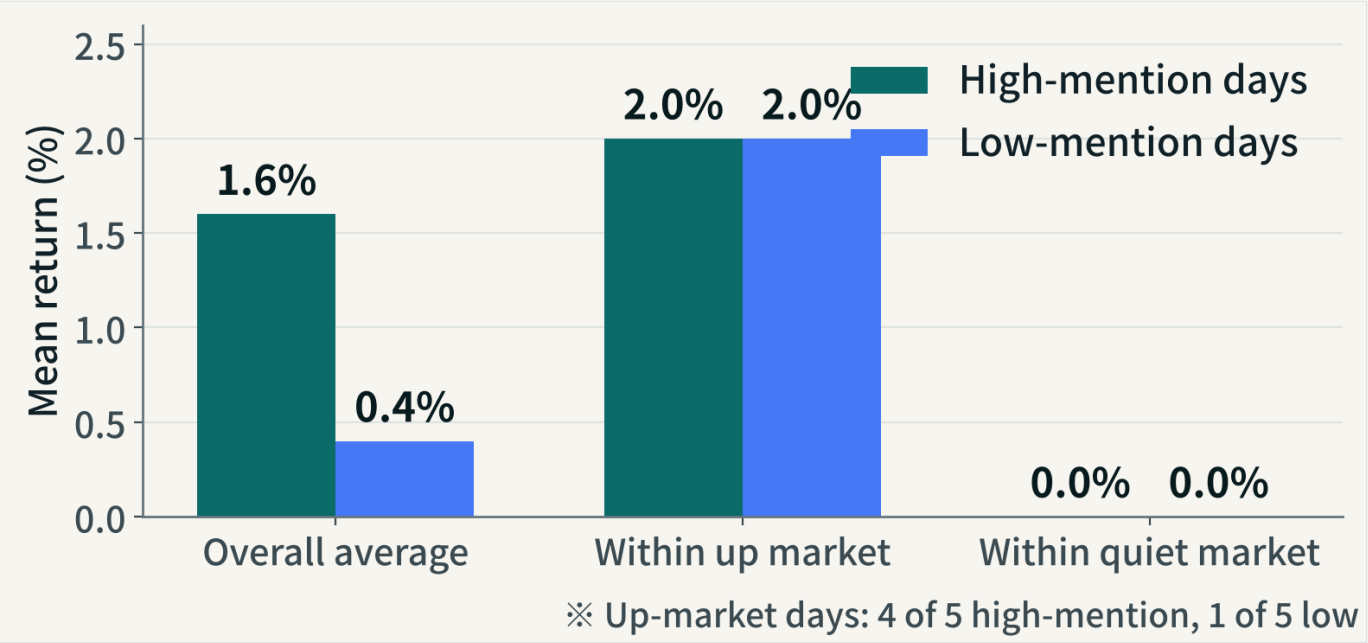
Correlation alone shows no causal effect

An overall gap can come from the market-regime mix

High-mention days 1.6%, low 0.4% — a 1.2%p gap

Within each regime

- In the bull regime, both groups earn 2%.
- In the quiet regime, both earn 0%.



The gap came from the share of bull days

Causes need a comparable counterfactual

Did texting customers increase purchases?

Randomized trial

Text one group only

- Raises comparability on average.
- Check sample size and compliance.

Not identical in finite samples

Observational inference

When trials are hard

- Random assignment is hard (e.g. rates).
- Requires no unobserved confounding.

More variables prove nothing

Intervention claims need matching evidence

SHAP splits a prediction fairly among features

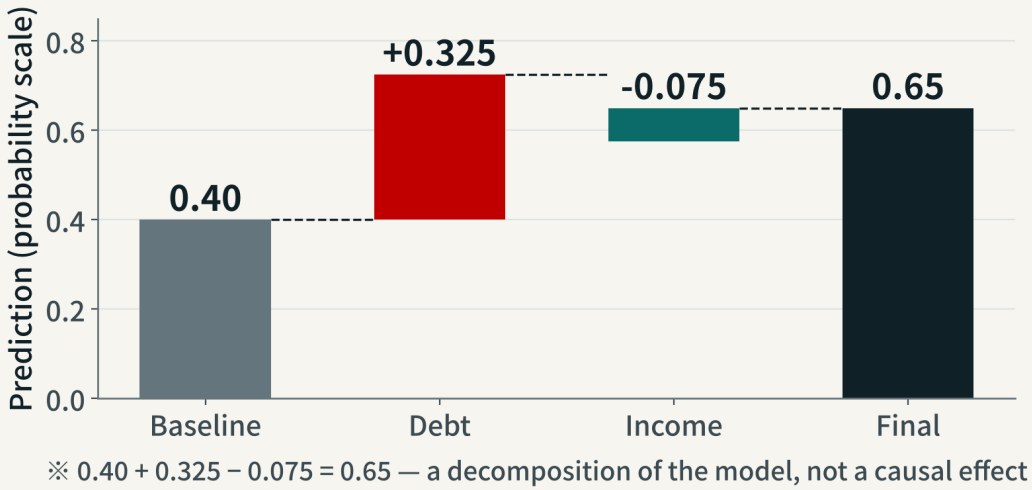
Base 0.40, debt only 0.70, income only 0.30, both 0.65 — average over feature orders

Shapley value

$$\phi_{\text{debt}} = \frac{(0.70 - 0.40) + (0.65 - 0.30)}{2} = 0.325$$

$$\phi_{\text{income}} = \frac{(0.30 - 0.40) + (0.65 - 0.70)}{2} = -0.075$$

$$\text{baseline} + \phi_{\text{debt}} + \phi_{\text{income}} = 0.40 + 0.325 - 0.075 = 0.65$$



※ 0.40 + 0.325 − 0.075 = 0.65 — a decomposition of the model, not a causal effect

Debt info pushed this model's prediction up; income pushed it down

Not a causal claim that cutting debt cuts risk by 32.5%p

Check four things when reading SHAP values

SHAP explains model math; it proves neither real causes nor social fairness

Causal?

- Model output, not cause.

Baseline

- Depends on background data.

Scale

- Log-odds are not %p.

Correlated features

- Depends on the split rule.

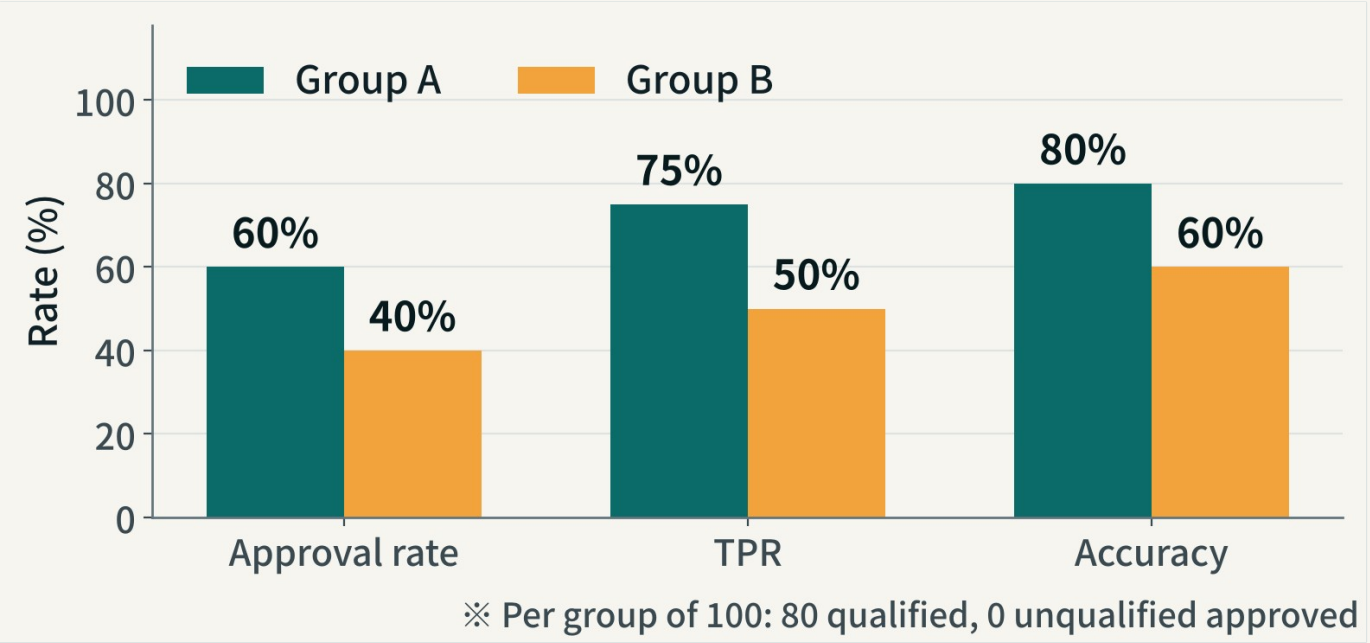
Fair math division and social fairness are different ideas

An overall average can hide group gaps

Two groups of 100, 80 qualified each; A approved 60 qualified, B 40

Denominators differ

- Approval rate: everyone.
- TPR: the qualified only.



A 25%p TPR gap appears

Demographic parity, equal opportunity and calibration differ and rarely hold together

A deleted sensitive column can live on in other variables

Even without a gender column, proxies may track that attribute

Proxy candidates

- Postcode, school and spending can stand in.

Also check

- Variable–group links, group error rates, and changes when dropped.

Limits

- SHAP alone cannot settle proxy issues.

Equal accuracy can hide group gaps in misses, false alarms, chances

03

Lessons from cases

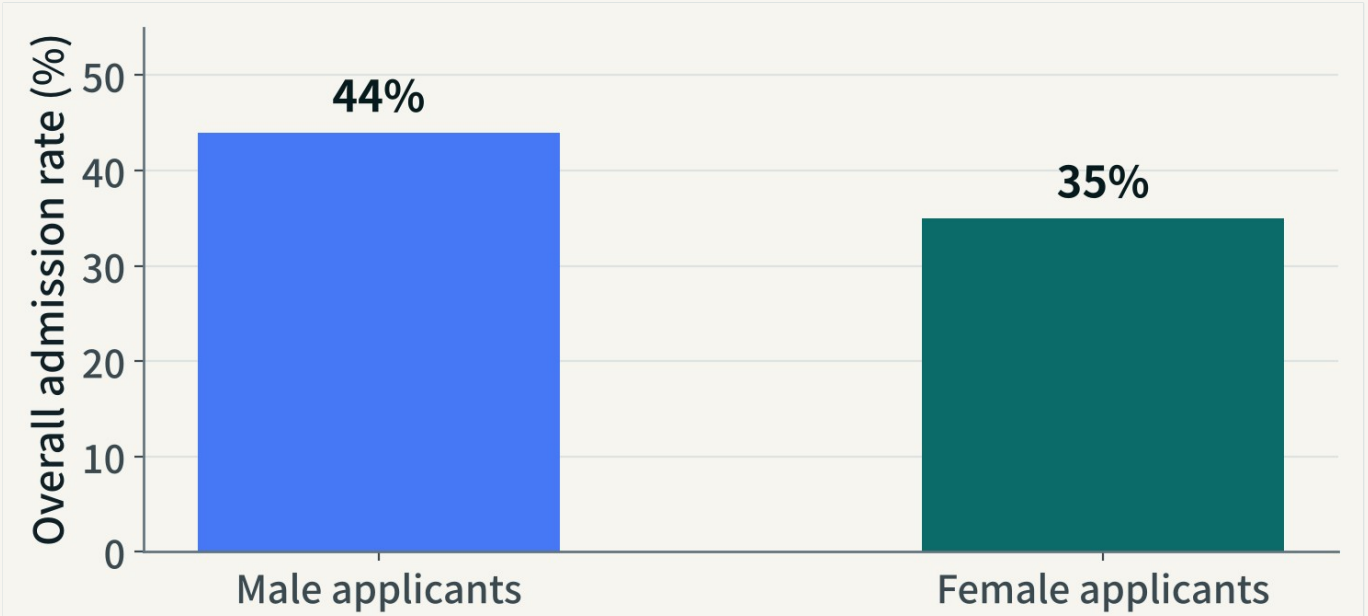
What averages and models can hide

An overall admission gap once came from department mix

UC Berkeley graduate admissions (1973): Simpson’s paradox

By department

- Women applied more to more competitive departments.
- Within departments, small or reversed gaps.



※ Bickel et al. (1975, Science). Split by department, most of the gap disappeared

Overall averages misplace the cause

Same structure as the main deck's ‘regime’ case

Dropping sensitive columns lets gaps in via other variables

Reported cases where proxies and explainability mattered

Hiring tool (2018 report)

- Amazon reportedly scrapped a tool trained on past resumes that penalized words like ‘women’s’.

Card limits (2019)

- NY DFS probed alleged gender gaps in Apple Card limits.
- In 2021 it found no fair-lending violation.

Lesson

- Explain group error rates and decision grounds.

Legal but unexplainable still loses trust

Fairness has several criteria, hard to meet at once

Three criteria on the main deck's loan case

Criterion	What is equal	Main deck case
Approval parity	Approval rate by group	A 60% vs. B 40%: differ
Equal opportunity	Approval among qualified (TPR)	A 75% vs. B 50%: differ
Calibration	Qualified share at a score	Needs score data to check

Pick and state the criterion by error costs and goals

04

Summary

Close the week with key Q&A

Prediction, explanation, causality need distinct evidence

One question, one answer for this week

Check question	Answer
Is a large SHAP value a real cause?	**No. It shows how the model used that feature.**
Same overall accuracy = fair?	**Gaps in misses, false alarms, opportunity may remain.**
Drop sensitive columns, solved?	**Proxies may remain; check errors by group.**
More variables prove causation?	**No. It needs identifying assumptions or experiments.**

Back intervention claims with fitting evidence

PREDICT.

EXPLAIN.

THEN ASK WHY.

Check groups and causes
hidden behind the average.